Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider two sources, X and Y, that emit symbols according to the following joint p.m.f.

$$\Pr\{X = i, Y = j\} = w_{i,j}, \tag{1}$$

   where the probability values $w_{i,j}$ are defined in the matrix below (i: rows, j: columns)

$$\mathbf{W} = \frac{1}{8} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

   (a) Compute the entropy of both sources, i.e., $H(X)$ and $H(Y)$.

   (b) Compute the joint entropy $H(X, Y)$

   (c) Design an Huffman code for the source X and compare the average code length with $H(X)$.

   (d) Design an Huffman code to jointly encode X and Y and compare the average code length with $H(X, Y)$.

2. **Question (5 pts)**

   (a) Illustrate the block diagram of an open loop vocoder.

   (b) Discuss the use of short and long term predictors.

3. **Question (5 pts)**

   (a) Illustrate the design of the compander-expander quantizer.

   (b) Discuss the coding efficiency with respect to the uniform quantizer and the Lloyd-Max quantizer.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a speech coding system

   - [T] [F] The number of prediction coefficients used in the short term predictor is approximately equal to 100.
   - [T] [F] Direct quantization of prediction coefficients might lead to instability at the decoder
   - [T] [F] The number of zeros of the whitening filter $A(z)$ depends on the number of formants below 400 Hz

   (b) Consider an image coding system

   - [T] [F] The 2D transform is invertible.
   - [T] [F] The coding efficiency increases when the block size is increased.
   - [T] [F] The complexity of the 2D DCT is $O(N^2 \log N)$
   - [T] [F] The color components in natural images (e.g., RGB components) are statistically independent.

   (c) Consider a video coding system.

   - [T] [F] The average MSE distortion of prediction residuals increases when the block size increases.
   - [T] [F] The average MSE distortion of prediction residuals of full search motion estimation is smaller than (or equal to) that of fast search.
   - [T] [F] Motion vectors of neighboring blocks are statistically dependent.

5. **Question (Matlab - 10pts)**

   Local image features are routinely used in several computer vision applications, including augmented reality (e.g Google Goggles). Their success is due to the ability of summarizing the input image in a way that is invariant to several transformations (scale, rotation, luminance, viewpoint etc.).

   You are requested to analyze the performance of lossless and lossy compression when applied to the feature set.

   (a) Load the file `feature_set.mat`. It contains a collection of 64-dimensional SURF local features extracted from a test image (each feature corresponds to one column in the feature set).

   (b) Lossless coding:

   - Assume the feature set as a *single* discrete memoryless source X, which emits symbols quantized with 8 bits. Compute the empirical entropy $H(X)$ (in bits / symbol) and the expected rate R (in bits / vector) for lossless coding a single feature vector.
   - Now assume that each of the 64 dimensions of the feature set as a different source, i.e., X_i ($i = 1 \dots 64$). Compute the empirical entropies $H(X_i)$ and the expected rate $R = \sum_{i=1}^{64} H(X_i)$.

   (c) Lossy coding: Encode the feature set with a PCM coding scheme. Use a uniform quantizer with granular region between -0.5 and 0.5 and with $M = \{2, 4, 8, 16, 32, 64, 128\}$ levels. Plot the rate-distortion curve.